

Credibility Scoring for Web Sources

Project 1 Report

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1. Introduction

A chatbot that answers questions is only as trustworthy as the sources it picks and cites. The issue is that its answers look identical whether the source is a peer-reviewed study, or a blog post written in five minutes by someone who has no understanding of the subject aside from what they've googled. This project was done to make that difference visible. When given a URL, the function `score_url()` returns a credibility score between 0 and 1, along with a plain-language explanation a reader can act on.

We started with the baseline provided by my professor, which combined a hand-written URL rule with a Claude judgment. With the help of Claude, we fixed seven of the twelve documented weaknesses, added a signal from real scholarly metadata (OpenAlex) and re-tuned how the layers are blended. Measured on 37 labelled URLs with the LLM layer on, the MAE (mean absolute error) fell from 0.117 to 0.060, and the share of URLs landing in the correct HIGH, MEDIUM, or LOW band rose from 81.1% to 94.6%. The rest of this report reviews related work (Section 2), describes the algorithm itself and why it was chosen (Section 3), reports the measured before-and-after results in tables (Section 4), discusses what still fails (Section 5), and concludes (Section 6).

2. Related Work

Early work on web credibility studied how people judge it. In a study of more than 2,500 participants rating live websites, Fogg et al. (2003) found that surface presentation weighed heavily; “design look” was the most frequently mentioned factor, in about 46% of the comments. Metzger (2007) surveys the skills, recommendations and cognitive models behind how users assess online information, treating credibility largely as a matter of user judgment and literacy. Both support scoring the source itself rather than how a page looks, which is also all a URL-based scorer can do.

Automated approaches, on the other hand, have mostly scored whole outlets. Baly et al. (2018) predict the factuality and political bias of news media from several features: the outlet's articles, its Wikipedia page, its Twitter account, the structure of its URL, and the web traffic it attracts, and each of those contributes to the score. The rule layer uses only URL structure, which is just one of the several signals used in that work. On the manual side, there is Wikipedia's community-maintained Perennial Sources list (Wikipedia contributors, n.d.) which rates frequently discussed sources by consensus into categories such as ‘generally reliable’, while stressing that it is not a complete list, nor a set of pre-approved sources. A list like this generally has a more defensible foundation than the hand-typed domain table that I inherited, but replacing it was one of the weaknesses I didn't address.

For scholarly sources, OpenAlex (Priem et al., 2022) is an open index of works, authors, venues and institutions with a free API. Its works endpoint returns fields I use directly: a retraction flag, a citation count, and the type of venue that hosts a work, such as a journal or an online repository. The paper itself describes the index and its venues; I confirmed the retraction and citation fields by querying the API (a retracted *Lancet* paper returned `is_retracted: true`, and a 2015 *Nature* article returned 85,205 citations).

Finally, language models themselves are increasingly being used as judges. Zheng et al. (2023) report that strong LLM judges reach over 80% agreement with human preferences, while also showing position, verbosity, and self-enhancement biases. The third layer, the LLM layer, is one of these judges, and here it is blended with the rule-based and metadata signals instead of being used alone. This project scores individual URLs and, when a DOI is present, uses facts about the paper itself rather than rating its credibility from just the domain alone.

3. Algorithm and Design Choices

3.1 The Baseline

The provided scorer has two layers. The rule layer inspects the URL string: it looks the domain up in a table of 25 hand-scored domains, falls back to a score for the top-level domain, adjusts for HTTPS and for path fragments such as `/sponsored/` or `/blog/`, and adds a bonus when the path contains a DOI. All of these are summed up into a single number. The second layer asks Claude (claude-opus-5) to judge the URL itself and then return a score with a one-sentence reason, and the two are blended as $0.6 \times \text{rules} + 0.4 \times \text{LLM}$. The baseline’s weaknesses are documented within its code. The ones we focused on were that it never reads the page, that additive scores cannot express an upper bound, and that the blend weight was never tested.

3.2 Ceilings

The first design idea we tackled was the ceiling. An additive score cannot say that a source must never exceed a certain value, yet some of the baseline failures are exactly of that kind. A sponsored article in a respected newspaper would inherit the newspaper’s reputation, and then there’s a DOI bonus lifting a preprint above a Wikipedia article. We represented these as “cap” signals: each rule can contribute an upper limit, and the final score is the smaller of the blended score and every active limit.

When figuring out when to apply the limits, we found that it was better to apply them after the LLM blend. For example, `blogs.nytimes.com` by itself, with just the rules, scored 0.55, while the model gives it 0.80. At the final weights, the blend would become 0.70, which crosses into the HIGH band. With the ceiling, the score stayed at 0.55. Without the post-blend ceiling, a generous model would quietly undo the rule fixes.

3.3 Fixes to the rule layer

Table 1 lists the weaknesses we addressed, the changes made, and the effect on individual URLs.

Weakness	Change	Effect
Preprints scored like peer-reviewed papers	Lower base scores for arXiv (0.75 to 0.65) and bioRxiv (0.70 to 0.55); added medRxiv, SSRN and ResearchSquare; no DOI bonus on preprint hosts, because a DOI there would say “registered”, not “reviewed”	bioRxiv went from 0.82 to 0.57 (label 0.50) arXiv 0.77 to 0.67 (label 0.65)
Subdomains inherit the parent’s score	Subdomains such as blogs., forums., opinion. on a known domain are capped at 0.55, with a small penalty on unknown domains; language subdomains such as en.wikipedia.org were left untouched	blogs.nytimes.com went from 0.82 to 0.55 forums.bbc.com went from 0.84 to 0.55
Any .edu scores high	Any /~name/ path was considered a personal page, and thus, capped at 0.50 on any host	web.mit.edu/~jdoe went from 0.84 to 0.50
Paid content scores middling; penalties stack	/sponsored/, /advertorial/ and /press-release/ were capped at 0.25; the total path penalty was floored at −0.30	nytimes.com/sponsored/... went from 0.62 to 0.25
Explanations were fragments joined by semicolons	The verdict comes first, followed by plain-language reasons from a 28-domain notes table, with any retraction listed first, and a “rough estimate” warning for unfamiliar sites	Example below

Table 1. Rule-layer fixes and their effect on URLs (rules only)

The ceiling values chosen, (0.55, 0.50 and 0.25) are stated assumptions rather than fitted quantities, and this is one of the limits discussed in Section 5. To illustrate the explanation change, the baseline had described arxiv.org as “‘arxiv.org’ is a domain we recognize; served over HTTPS”. Now, instead, the new text reads: “Medium credibility (0.67). arxiv.org is a preprint server, so papers are posted without peer review (base score 0.65). It uses HTTPS, a very small plus since scam sites can get certificates too.”

3.4 Real metadata from OpenAlex

The rule layer judges a string, so when the URL contains a DOI, we can instead ask about the work itself rather than just the URL. The scorer extracts the DOI from the URL path and queries

OpenAlex, which needs no API key (unlike the LLM layer), then turns the response into signals. A retracted paper is given a ceiling of 0.10, applied both before and after the LLM blend, and a work held in an indexed journal earns a small bonus of 0.08, while one that’s held only in a repository is capped at 0.65, because a repository implies no peer review. Citations add a bonus of at most 0.07, which is scaled by the log of the count, because citations are weak and quite frankly, noisy evidence since a bad paper can be cited for the purpose of refuting it.

The lookup itself is treated as optional. It has a five-second timeout, and then results are cached, and any failure whether it was offline, had an unknown DOI, or timed out, falls back to the rules and prints a single warning, so the application never breaks. This is better than just a longer lookup table because it uses facts about the paper rather than guesses based on its domain. The clearest case of this is the retracted *Lancet* paper. The rules scored 0.72 because of the .org along with a respectable DOI, even though the LLM alone scored 0.20, so the blend landed near 0.5. Now with the retraction flag, the rule score drops to 0.10, and the final score is capped there.

3.5 The Blend

The baseline itself weighed the rules at 0.60 and the model at 0.40, with no evidence to explain why this was chosen as the split. We swept the rule weight from 0 to 1 and measured the error at each value (shown in Section 4.3), then chose 0.4 for the rules and 0.6 for the model.

4. Evaluation

4.1 The Setup

I evaluated 37 labelled URLs: the 24 provided with the project, plus 13 that I chose with the help of Claude. The labels are judgments, not ground truth, and I say more about that in Section 5. I report three measures: MAE (mean absolute error) against the label, band accuracy (whether the HIGH, MEDIUM or LOW chip is correct, with HIGH at 0.70 and above, LOW below 0.40, and MEDIUM in between), and the worst single error. The LLM layer used claude-opus-5.

4.2 Main Results

Table 2 compares my configuration against the original scorers:

Configuration	Original	Improved
Rules Only	0.184 – 54.1% – 0.67	0.151 – 67.6% – 0.67
Rules + OpenAlex	N/A	0.126 – 73.0% – 0.44
Rules + LLM	0.117 – 81.1% – 0.54	0.072 – 89.2% – 0.42
Rules + LLM + OpenAlex	N/A	0.060 – 94.6% – 0.15

Table 2. Results on the thirty-seven labelled URLs (MAE – band accuracy – worst error)

With the LLM layer enabled, the entire pipeline (all three layers) cuts MAE roughly in half, dropping from 0.117 to 0.060, and band accuracy rises by 13.5 percentage points, jumping from 81.1% to 94.6%. The worst error also falls from 0.54 to 0.15. On the rules-only side, the drop in MAE from 0.184 to 0.151 came from the preprint, subdomain, promotional and personal-page fixes. The LLM layer on its own brings the score down from 0.184 to 0.117, which is why the errors are lower in every row where it is enabled. The changes to the scorer bring that down further, from 0.117 to 0.072, and part of that gain comes from re-tuning the blend weights, which only matters when the LLM layer is blended in. Both LLM rows in the improved column use the same final weights, so the drop in worst error from 0.42 to 0.15 and the drop in MAE from 0.072 to 0.060 come from OpenAlex alone, mainly because it catches the retracted paper.

4.3 Choosing Blend Weight

To choose the weights of the rules and the LLM layer without paying for anything because I am cheap sometimes, each URL's rule score and model score was collected, and then blended. Table 3 shows selected settings on the 36 URLs that existed at the time. The 37th URL, blogs.nytimes.com, was added after this sweep to test the post-blend ceiling, so Table 3 uses the first 36.

Rule Weight	MAE	Bands	Worst Error
0.00 (Model Only)	0.071	86.1%	0.20
0.20 (Lowest MAE)	0.050	91.7%	0.15
0.40 (Chosen)	0.061	94.4%	0.16
0.60 (Original)	0.077	88.9%	0.24
1.00 (Rules Only)	0.130	72.2%	0.44

Table 3. Effect of the rule weights on the 36 URLs

Table 3 shows that 0.20 has the lowest error, so I should explain why I didn't pick it. The error curve is pretty flat between 0.15 and 0.30, within 0.005 of its minimum, so the exact value doesn't matter much in the grand scheme of things. Leave-one-out cross-validation chose 0.20 on every fold, and tuning on the original 24 URLs and then testing on the ones I added favored 0.20. I still went with 0.40, because it one; gives the best band accuracy, and it keeps the rules as a real part of the score instead of seeing it as just an afterthought. The table also shows that the model on its own beats the rules on their own, so the rules had earned their place in the combination and through the ceilings, not by being accurate alone. Moving from 0.6 to 0.4 lowered the MAE from 0.077 to 0.062 on the 37 URLs, and applying the ceilings after the blend lowered it further to 0.058.

4.4 Live Check and Robustness

To perform a qualitative check, I asked the running application, "Is there a link between the MMR vaccine and autism?" It returned six sources, each with a score: a Wikipedia article on the Lancet report scoring a 0.69, immunize.org scoring at 0.79, followed by PolitiFact, which scored a 0.69 as well, then the University of Minnesota's CIDRAP, scoring a 0.89, followed by the New

England Journal of Medicine, which was scored at 0.99, and finally the Children’s Hospital of Philadelphia, which scored a 0.91. The ordering made sense: the peer-reviewed journal is on top, and the Wikipedia article and factchecker both sit in the middle. Now again, I would like to preface that this was one question that was asked, so I draw no conclusions from it.

The test itself has 77 checks, all of which can be performed offline, and this was achieved through the help of Claude. The model is faked, and the OpenAlex responses are injected, and it was cool putting that together. But I digress, a test of 43 unusual inputs, such as None, lists, bytes, IPv6 hosts, as well as 100,000-character length URLs and null bytes, found one defect in the original code: lists and dictionaries raised an error because the cache lookup ran before the input check. We fixed it by moving the check first. to prevent it from erroring out, which prevented sites like nytimes.com.evil.com, and others like it, to be scored as unknown sites and thus, do not inherit the real publisher’s score.

5. What Still Fails

The remaining errors are concentrated in plain domains that do not have a DOI. In these cases, the rules don’t know anything, and the LLM is the only informative signal. Table 4 lists the five largest misses in the final configuration.

Label	Score	URL	Why
0.08	0.23	naturalnews.com	Rules give an unknown .com 0.52, but the model says 0.03, so the blend is pulled up as a result
0.93	0.79	jamanetwork.com	The rules do not know JAMA, which is an automatic 0.52, but the model says 0.97, and there is no DOI to check with OpenAlex
0.35	0.48	seekingalpha.com	The model itself says 0.45, but there’s a band miss
0.65	0.75	arxiv.org	The model says 0.80, but again, there’s a band miss
0.88	0.78	who.int	Same as JAMA above

Table 4. The Largest Remaining Errors in the Final Configuration

Only two of the thirty-seven URLs landed in the wrong band, arXiv and Seeking Alpha. and there are several limits affecting how far these results can be trusted.

The main one being, again, the labels are my own judgment, and some of the 13 URLs were chosen to show specific fixes, such as the retracted paper, a blog subdomain, a personal page, a medical preprint and blogs.nytimes.com, so improvement on those specific things are built directly into the scorer. The fairer held-out cases are the plain domains, like the CDC, Mayo Clinic, the Guardian, TechCrunch, a company blog and [naturalnews](https://naturalnews.com). Second, the blend weights, both the ceilings and several base scores were chosen after seeing the labels. With 37 URLs, one run each and no confidence intervals, they are fitted to my data, and model outputs vary slightly between calls as shown in the original scorers model, which varied between 0.086, 0.087 and 0.088 across three runs, so differences below 0.005 are noise. Third, the post-blend ceiling's benefit shows up mostly on a URL I added to test it; on the labels provided it changes nothing.

Then comes in OpenAlex, which is only helpful when the URL contains a DOI. One of the provided URLs, the PNAS article, has a DOI that doesn't exist, so OpenAlex returns not found, and the identifier form I tried for arXiv didn't resolve. The ceilings also override the model by design, so a good personal page or blog post would generally be stuck at 0.50 or 0.55, even if it's knowledgeable or cites vetted sources. On top of that, the `/~` rule misses `people/` pages on `.edu` servers, which cannot be told apart from departmental pages by URL alone. I also tested a 0.97 cap in the rule layer to stop a known journal from reaching exactly 1.00. It changed the MAE by only about 0.0003, so I left it out, since it didn't make a marginal enough difference to include it or not, and technically nothing can be perfect.

Five of the twelve documented weaknesses remain unaddressed: reading the page itself, a sourced domain table, learning the weights by regression, calibrating the score, as well as reporting uncertainty. I think the most valuable next step would be to read the page content, such as the author, citations, the date, if there's a corrections policy or not, things of that nature, since that is the signal the URL-only rule layer lacks, followed by calibrating the score against labelled data so that if something is scored as, say, 0.7, it truly is a 0.7.

6. Conclusion

The baseline credibility scorer was improved by adding upper limits that additive rules alone couldn't express, by using real retraction and venue data whenever a DOI was available, and by measuring the blend between rules and an LLM instead of guessing what rule weight would fit better. On 37 labelled URLs, the final configuration lowered the MAE from 0.117 to 0.060 and raised band accuracy from 81.1% to 94.6%. As mentioned above, to further improve the scorer, the next step is to read the entire page and its contents, because the largest errors come from domains that the rules themselves cannot gauge effectively.

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